

- 10** Hydroxyproline is synthesised by addition of an –OH group to the R-group of the amino acid proline.

Hydroxyproline is a major component of collagen and has an important role in increasing the stability of its structure.

Which statement explains why the addition of an –OH group to proline could increase the stability of collagen?

- A** It strengthens hydrogen bonding between the R-groups of adjacent polypeptide chains, resulting in a tertiary structure that is more resistant to heat denaturation.
- B** It increases the number of sites available for the formation of hydrogen bonds within the secondary structure of collagen, resulting in more stable alpha helices.
- C** It increases the formation of hydrogen bonds between R-groups and water molecules, which help to hold the chains of the collagen triple helix together by strengthening hydrophilic interactions.
- D** It strengthens the quaternary structure of collagen by providing more sites for hydrogen bonding between the R-groups of distantly separated amino acids within the same polypeptide chain.